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Unit – 1 : Session – 9 : SLO – 1 | Functional Programming Concepts

■ Activity (Solutions)

- 1) Pure function with `Function<T,R>`: `java.util.function.Function<Integer,Integer> square = x -> x * x;`
`System.out.println(square.apply(6));` // 36 — pure: same input always gives same output, no side effects.
- 2) Higher-order function = a function that takes a function as an argument or returns one. Java example:
`static int applyTwice(IntUnaryOperator f, int v){ return f.applyAsInt(f.applyAsInt(v)); }` `applyTwice(x -> x + 1, 5)`
→ 7. Also `list.forEach(System.out::println)`.
- 3) Immutability with `final`: `final int x = 10;` // `x = 20;` // compile error — reassignment forbidden. `final List<String> names = List.of("a","b");` // `names.add("c")` throws `UnsupportedOperationException` — state cannot change after creation, a core functional principle.